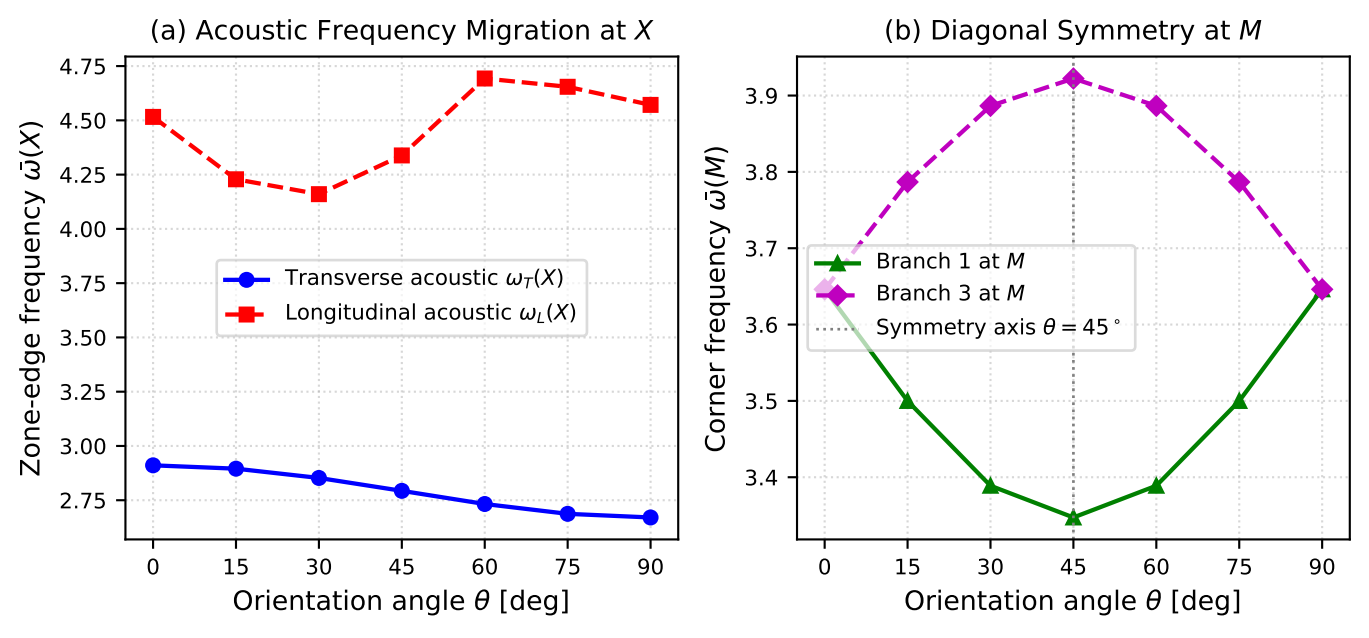
(a) Acoustic Frequency Migration at X
Zone-edge frequency $\bar{\omega}(X)$
4.75
4.50
4.25
4.00
3.75
3.50
3.25
3.00
2.75
Transverse acoustic $\omega_T(X)$
Longitudinal acoustic $\omega_L(X)$
0
15
30
45
60
75
90
Orientation angle θ [deg]
(b) Diagonal Symmetry at M
Corner frequency $\bar{\omega}(M)$
3.9
3.8
3.7
3.6
3.5
3.4
Branch 1 at M
Branch 3 at M
Symmetry axis $\theta = 45^\circ$
0
15
30
45
60
75
90
Orientation angle θ [deg]